

Sara McGhee

SOFTWARE ENGINEER · AI & RETRIEVAL SYSTEMS

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SUMMARY

Software engineer who puts AI into systems that already exist. Four years delivering insurance products across the quote-to-claim pipeline — mainframe DB2 through Kafka microservices to React — and **two-time company hackathon winner** for enterprise retrieval systems on Azure. **Voted team MVP three years running** (2023–2025).

EXPERIENCE

Software Engineer II

2022 – Present

Unum · Chattanooga, TN

- **Deliver new insurance products across the quote-to-claim pipeline** — quoting, policy issue, billing and claims — moving between stacks as each layer requires: Java on Linux, C# and .NET, React and JavaScript, MSSQL, and DB2 on the mainframe.
- **Built microservices integrating third-party vendors**, event-driven over **Kafka**, with outbound calls brokered through a **Layer7 API gateway**.
- **Extended a claims system on MSSQL** and built data-warehouse batch jobs supporting downstream reporting.
- **Won the 2025 company hackathon**. Built the ingestion pipeline and retrieval system for an onboarding knowledge assistant — Word and Excel documents normalized to JSON via Power Platform, indexed in **Azure AI Search** as a hybrid vector-plus-keyword index over **Blob Storage**, answered by **Azure AI Foundry** with required citations.
- **Won the 2024 company hackathon**. Built the backend for conversational retrieval over 10,000+ enterprise business rules — Pinecone vector index, embedding pipeline, and model tuning.

Career break — family

2014 – 2021

Returned to engineering through a full-stack development bootcamp.

Software Engineer

2009 – 2014

Symvionics, Inc. · Pensacola, FL

- Built and debugged large distributed systems for U.S. Navy training — radar simulations and carrier air traffic control simulators.
- Owned requirements, design, test development, verification and certification against government specifications; collaborated on embedded processing systems.

SELECTED PROJECTS

Legacy Automation Broker — C# / .NET 8 · github.com/sarajmcghee/LegacyAutomation

An agent driving a legacy ASP.NET WebForms application through Edge IE mode, grounded in DOM state rather than screenshots so every action is verified before the next is allowed.

Birdsong evaluation audit — Python · [scikit-learn](https://scikit-learn.org/)

Built image and acoustic classifiers on CUB-200-2011, then found my own 97.8% baseline was measuring recording identity rather than species — a 1-NN lookup beat the trained classifier. Published the corrected ~89% and the method, not the flattering number.

SKILLS

AI / ML	Azure AI Foundry, Azure AI Search, RAG and hybrid retrieval, embeddings, Pinecone, PyTorch, scikit-learn, model evaluation
LANGUAGES	C#, Java, Python, JavaScript, TypeScript, SQL
DATA & PLATFORM	MSSQL, DB2 (mainframe), data warehousing and batch ETL, Kafka, REST APIs, Layer7 API Gateway, .NET 8, Linux, Azure, Power Platform, React, Node

EDUCATION & CERTIFICATIONS

IN PROGRESS	M.S. Computer Science, AI/ML focus — Colorado State University Global
2022	Full-Stack Web Development — NuCamp Coding Bootcamp
2009	B.S. Computer Science, minor in Mathematics — University of Nevada, Las Vegas
CERTIFIED	Microsoft Azure Fundamentals (AZ-900) · GitHub Foundations · OpenText Exstream Designer